5-Day Workshop Plan

Day 1: Python Basics, Lists and Strings

Topics

- Basic Introduction
- Conditions: if, if-else, elif, nested if, ternary expression
- Iteration: for, while, range(), nested loops, loop with else, break, continue, pass
- Lists: introduction, indexing (positive and negative), slicing with step, nested lists, list operations (+, *, in)
- List methods and functions (15): append(), extend(), insert(), remove(), pop(), clear(), index(), count(), sort(), reverse(), copy(), len(), sum(), min(), max()
- List comprehensions, with condition and nested; sorted(), enumerate(), zip()
- Strings: literals (single, double, triple quotes, raw strings), escape sequences, indexing, slicing, immutability, operations (+, *, in)
- String methods (18): upper(), lower(), title(), capitalize(), swapcase(), strip(), lstrip(), rstrip(), split(), join(), replace(), find(), index(), count(), startswith(), endswith(), isdigit(), isalpha()
- Other string methods: center(), zfill(), isalnum(), isspace(), partition(); string formatting

Day 2: Tuples, Sets, Dictionaries & Functions

Topics

- Tuple methods and functions (12): count(), index(), len(), min(), max(), sum(), sorted(), tuple(), any(), all(), enumerate(), reversed()
- Set methods (15): add(), update(), remove(), discard(), pop(), clear(), copy(), union(), intersection(), difference(), symmetric_difference(), issubset(), issuperset(), isdisjoint(), len()
- Dictionary methods (15): get(), keys(), values(), items(), update(), pop(), popitem(), setdefault(), clear(), copy(), fromkeys(), del, len(), in, sorted()
- Dictionary comprehensions and sorting a dictionary by key or value
- Functions: defining and calling, return statement, returning multiple values, docstrings
- Arguments: positional, keyword, default, variable-length (*args and **kwargs), passing lists and dictionaries
- Scope: local, global, nonlocal, LEGB rule; nested functions
- Lambda functions: syntax, lambda with conditions, lambda with sorted(), min(), max()
- map(): applying a function to every item, with lambda and with multiple iterables
- filter(): selecting items by condition, with lambda
- reduce() from functools: syntax, sum, product and maximum using reduce(), comparison with loops
- Combining map(), filter() and reduce() in one pipeline; list comprehension vs map/filter

Day 3: Recursion, Modules and Packages

Topics

- Recursion: definition, base case and recursive case, how the call stack works, stack overflow and recursion limit (sys.setrecursionlimit)
- Recursion vs iteration: readability, speed and memory
- Recursive examples: factorial, Fibonacci, sum of natural numbers, sum of digits, power, GCD, string reversal, palindrome check
- Recursion on lists and strings: sum of a list, finding the maximum, counting characters
- Binary search using recursion; Tower of Hanoi
- Improving recursion with memoization (dictionary cache and functools.lru_cache)
- Inbuilt functions: input(), eval(), map(), filter(), len(), range(), zip(), enumerate(), abs(), round(), type()
- Modules and packages: creating and importing user-defined modules, import styles (import, from-import, as), __name__ == "__main__", package structure with __init__.py
- math module (12): sqrt(), pow(), ceil(), floor(), factorial(), gcd(), fabs(), log(), sin(), cos(), pi, e
- time module (8): time(), sleep(), ctime(), localtime(), gmtime(), strftime(), mktime(), perf_counter()
- OS module (12): getcwd(), listdir(), mkdir(), makedirs(), rmdir(), remove(), rename(), path.exists(), path.join(), path.isfile(), path.isdir(), path.getsize()
- random module: random(), randint(), choice(), shuffle(), sample(), seed()
- datetime module: date, time, datetime, timedelta, formatting and date arithmetic
- Introduction to numpy (array, arange, reshape, mean, sum) and scipy

Day 4: Exceptions and Files

Topics

- Errors vs exceptions; common exceptions: ZeroDivisionError, ValueError, TypeError, IndexError, KeyError, FileNotFoundError
- try-except, multiple except blocks, catching several exceptions, else clause, finally
- raise statement, assert statement, User Defined Exceptions (custom exception classes)
- File handling: introduction, open() function and its syntax, close() method, using the with statement
- File modes: r, w, a, r+, rb, wb
- File methods (10): read(), readline(), readlines(), write(), writelines(), seek(), tell(), flush(), close(), truncate()
- Working with text files, binary files and CSV-style data; checking files with the OS module

Day 5: Regex and Project

Topics

- Regular expressions: the re module, raw strings, metacharacters (. ^ $ * + ? [] {} | \), character classes (\d \w \s)
- match(), search(), findall(), finditer(), split(), replace() using sub()
- Groups and compile(): extracting parts of a match

- Database Connectivity Overview
- ***Project***

Sample Project ideas

- Contact book: add, search, update and delete contacts, saved in a file
- Student grade analyzer: marks stored in lists and dictionaries, with average, topper and grade using lambda, map, filter and reduce
- Quiz game: random questions, score calculation and score history saved in a file
- Expense tracker: record expenses by category, show totals, and save to a file
- Email and log validator: check emails and phone numbers with regex and report errors
- ATM or library management simulation using functions and exception handling

Note: Please give students practice questions every day based on the topics covered that day, make sure every student completes them, and let students discuss their doubts if required.